

Pendulum Modeling TF

EGR 345

GVSU

Outline

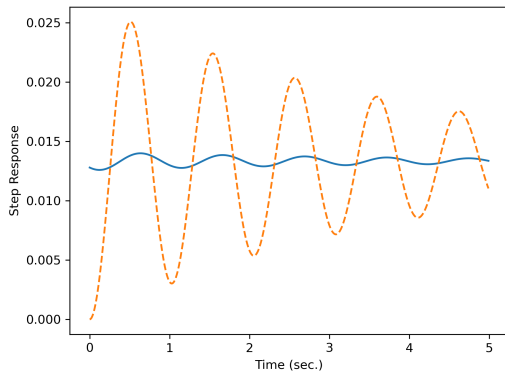
- debugging my MSD example from last lecture
- pendulum modeling
 - displacement input TF

Teaching Philosophy Issue

- I don't script my classes and work out all my examples ahead of time...
 - bug or feature?

Step Response Issue from Last Time

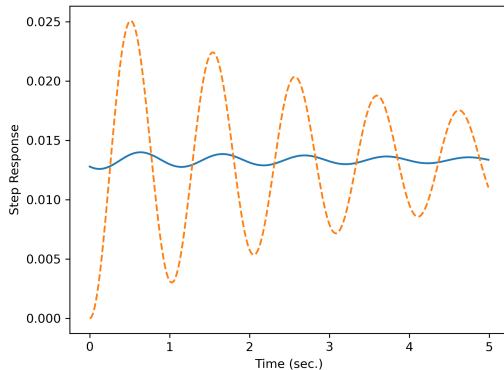
- don't blurt out the answer yet
- I see two issues with this graph, one quantitative and one qualitative



Step Response Issue Continued

Known form:

$$y(t) = C_1 \cdot 1(t) + e^{-at} (C_2 \sin(\omega_d t) + C_3 \cos(\omega_d t))$$



Pendulum Modeling

Is a pendulum basically the same as a mass/spring/damper system?

- what did you learn in last week's lab?

- initial condition response
- impulse response from tap/whack

} kind of fits ok-ish

- what about voltage input to the motor on the cart?



Two Approaches to Experimental System Identification

→ finding a system model from experimental data

1. Time Domain

- step or impulse response

or you need to guess or find TF before you fit the data

2. Frequency Domain

→ sinusoidal input

- 1 freq. or a range of frequencies

→ Bode Plots

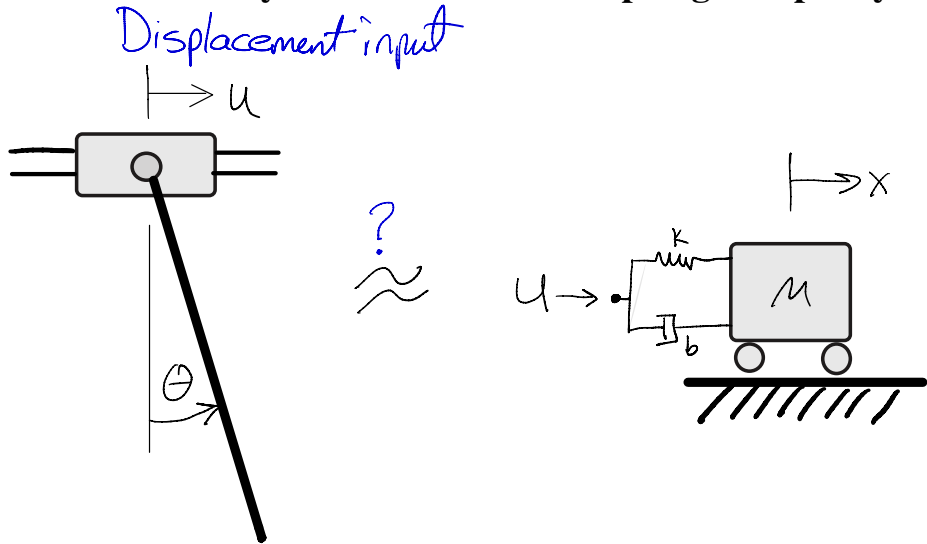
or Bode plot points to TF

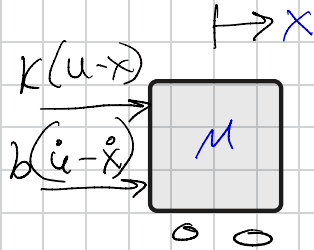
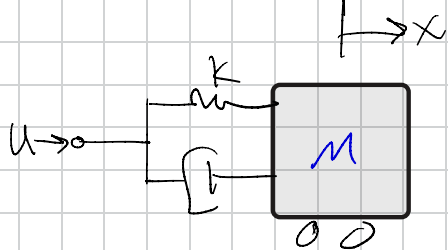
Lab 3

viscous damping $\neq$ dry friction



Is a pendulum basically the same as a mass/spring/damper system?





$$m\ddot{x} = k(u-x) + b(\dot{u} - \dot{x})$$

$$m\ddot{x} + b\dot{x} + kx = ku + b\dot{u}$$

$$\mathcal{L}[m\ddot{x} + b\dot{x} + kx = Ku + b\dot{u}]$$

$$(ms^2 + bs + k)X(s) = (k + bs)U(s)$$

$$G(s) = \frac{X(s)}{U(s)} = \frac{k + bs}{ms^2 + bs + k}$$

$$G(s) = \frac{X(s)}{U(s)} = \frac{K + bs}{ms^2 + bs + K}$$

Find the unit step response

$$u(t) = 1(t) \quad u(s) = \frac{1}{s}$$

$$X(s) = \frac{1}{s} \cdot \left(\frac{k + bs}{ms^2 + bs + k} \right) \frac{1/m}{1/m}$$

$$X(s) = \frac{1}{s} \cdot \frac{\omega_n^2 + 2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

$$= \frac{C_1}{s} + \frac{C_2\omega + C_3(s + a)}{(s + a)^2 + \omega^2}$$

$$X(s) = \frac{1}{s} \cdot \frac{\omega_n^2 + 2\zeta\omega_n s}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \frac{C_1}{s} + \frac{C_2\omega + C_3(s+a)}{(s+a)^2 + \omega^2}$$

$$a = \zeta\omega_n \quad \omega = \omega_n \sqrt{1 - \zeta^2}$$

$$\omega_n^2 + 2\zeta\omega_n s = C_1(s^2 + 2\zeta\omega_n s + \omega_n^2) + C_2s\omega + C_3s(s+a)$$

$$\omega_n^2 + 2\zeta\omega_n s = \underline{C_1 s^2} + \underline{C_1 2\zeta\omega_n s} + \underline{C_1 \omega_n^2} + \underline{C_2 s\omega} + \underline{C_3 s^2} + \underline{C_3 sa}$$

$$\underline{\omega_n^2} + \underline{2\zeta\omega_n s} = \underline{C_1 s^2} + \underline{C_1 2\zeta\omega_n s} + \underline{C_1 \omega_n^2} + \underline{C_2 s \omega} + \underline{C_3 s^2} + \underline{C_3 s \omega}$$

$$s^0: \omega_n^2 = C_1 \omega_n^2 \rightarrow C_1 = 1$$

$$s^1: 2\zeta\omega_n s = \cancel{C_1} 2\zeta\omega_n s + \cancel{C_2} s \omega + \cancel{C_3} s \omega$$

$\begin{matrix} -1 & \zeta\omega_n \\ \uparrow & \uparrow \end{matrix}$

$$2\zeta\omega_n = \zeta\omega_n + C_2 \omega \rightarrow C_2 = \frac{\zeta\omega_n}{\omega} = \frac{\zeta\omega_n}{\omega_n \sqrt{1-s^2}}$$

$$s^2: 0 = \cancel{C_1} s^2 + \cancel{C_3} s^2 \rightarrow C_3 = -C_1$$

$$x(t) = C_1 I(t) + e^{-\zeta \omega_n t} (C_2 \sin(\omega_d t) + C_3 \cos(\omega_d t))$$

Tangent: Overdamped 2nd order system

$$X(s) = \frac{1}{s} \cdot \frac{25}{s^2 + 15s + 25} = \frac{25}{s(s+1.91)(s+13.09)}$$

$$s^2 = \frac{-15 \pm \sqrt{15^2 - 4 \cdot 25}}{2}$$

$$p_1 = -1.91$$

$$p_2 = -13.09$$

$$X(s) = \frac{25}{s(s+1.9)(s+13.09)} = \frac{C_1}{s} + \frac{C_2}{s+1.9} + \frac{C_3}{s+13.09}$$

$$25 = C_1(s^2 + 15s + 25) + C_2 s(s+13.09) + C_3 s(s+1.9)$$

$$\underline{25} = \underline{C_1 s^2} + \underline{15 C_1 s} + \underline{25 C_1} + \underline{C_2 s^2} + \underline{13.09 C_2 s} + \underline{C_3 s^2} + \underline{C_3 1.9 s}$$

$$25 = \underline{C_1 s^2} + \underline{15 C_1 s} + \underline{25 C_1} + \underline{C_2 s^2} + \underline{13.09 C_2 s} + \underline{C_3 s^2} + \underline{C_3 1.915}$$

$$s^0: 25 = 25 C_1 \rightarrow C_1 = 1$$

$$s^1: 0 = \cancel{15 C_1} + 13.09 C_2 + 1.91 C_3 \rightarrow 13.09 C_2 + 1.91 C_3 = -15$$

$$s^2: 0 = C_1 + C_2 + C_3 \longrightarrow C_2 + C_3 = -1$$

$$C_2 = -1.171 \quad C_3 = 0.171$$

$$x(t) = C_1 \cdot 1(t) + C_2 e^{-19.4t} + C_3 e^{-13.9t}$$

→ overdamped systems
don't oscillate!